Experiment 17:. Random Sampling and Sampling Distribution

CODE:

```
import numpy as np

import matplotlib.pyplot as plt

population_mean = 50

population_std = 10

population_size = 100000

population = np.random.normal(population_mean, population_std,
population_size)

sample_sizes = [30, 50, 100] # different sample sizes to consider

num_samples = 1000 # number of samples for each sample size
```



```
sample_means = {}

for size in sample_sizes:

sample_means[size] = []

for _ in range(num_samples):

sample = np.random.choice(population, size=size, replace=False)

sample_means[size].append(np.mean(sample))

plt.figure(figsize=(12, 8))

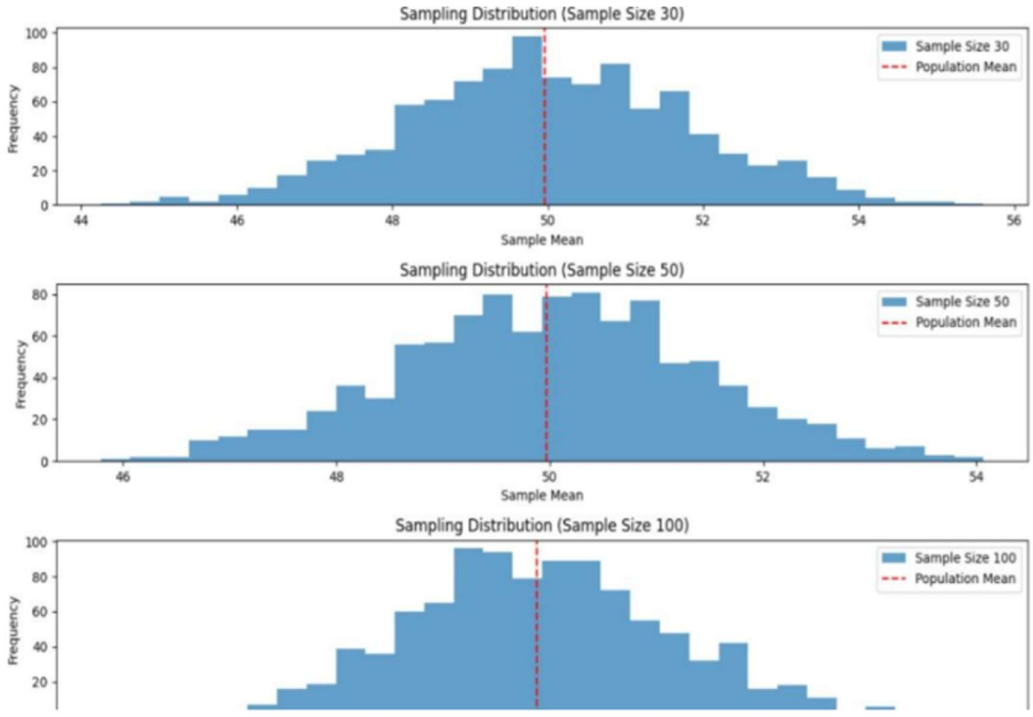
for i, size in enumerate(sample_sizes):

plt.subplot(len(sample_sizes), 1, i+1)

plt.hist(sample_means[size], bins=30, alpha=0.7, label=f'Sample Size
{size}')
```

```
plt.axvline(np.mean(population), color='red', linestyle='dashed',
linewidth=1.5, label='Population
Mean')
plt.title(f'Sampling Distribution (Sample Size {size})')
plt.xlabel('Sample Mean')
plt.ylabel('Frequency')
plt.legend()
plt.tight_layout()
plt.show()
```

OUTPUT



RESULT: Hence Random Sampling and Sampling Distribution hs been successfully verified.